

PERSONAL INFORMATION	Janaka Brahmanage Contact: (+65) 84517953 — janakat.2022@phdcs.smu.edu.sg Website: janakact.github.io LinkedIn: janakachathuranga	
RESEARCH INTERESTS	Reinforcement Learning, Imitation Learning, Large Language Models, Constraint Satisfaction and Optimization, Functional Programming, Category Theory	
EDUCATION	School of Computing and Information Systems, Singapore Management University , Singapore. <i>PhD Candidate</i> · Supervisor: Prof. Akshat Kumar · Research Topic: Constrained Reinforcement Learning, Imitation Learning	Aug, 2022 - Present
	University of Moratuwa , Sri Lanka. <i>B.Sc. Eng (Hons.) in Computer Science & Engineering</i> · Class: First Class · Cumulative GPA: 3.99/4.2 · Rank: 17th (out of 125 Computer Science & Engineering graduates) · Electives: Theory of Computing, Programming Languages, Intelligent Systems, Image Processing, Formal Methods in Software Engineering, Digital Photography, Advanced Algorithms, Concurrent Programming, Machine Learning, Data Mining & Information Retrieval, Distributed Systems, Compiler Design, Bioinformatics	Jan, 2014 - Jan, 2018
	Royal College , Colombo, Sri Lanka. <i>General Certificate of Education Advanced Level in Physical Science stream (University Entrance)</i> · Rank: 42nd (out of ~40,000) · Rank: 5th in Colombo district	Jun, 2004 - Aug, 2012
EXPERIENCE	Huawei Singapore Research Centre , 9 North Buona Vista Dr, The Metropolis Tower 1, Singapore. <i>Research Intern</i> Develop algorithms for agentic search and chat assistants.	May, 2026 - Present
	School of Computing and Information Systems Singapore Management University , 81 Victoria Street, Singapore. <i>Research Engineer</i> Develop algorithms for bus frequency optimization and perform data mining on Singapore's public transportation system.	Nov, 2020 - July, 2022
	Stack Technologies (Pvt) Ltd. , 187 Ward Pl, Colombo 01, Sri Lanka. <i>Co-founder / Software Engineer</i> Responsible for software design, development, and leading the development team.	Oct, 2019 - Nov, 2021
	Hybriteq Solutions (Pvt) Ltd. , 430/5, Thaladena, Malambe, Sri Lanka. <i>Co-founder / Software Engineer</i> Responsible for software design, development, and leading the development team.	Aug, 2018 - Oct, 2019
	WSO2 Inc. , 20 Palm Grove, Colombo, Sri Lanka. <i>Engineering Intern</i> Successfully completed a software engineering internship of six months.	Aug-Dec, 2016

PUBLICATIONS

Full Papers

2026, Choudhury A.*, **Brahmanage, J. C.***, Kumar, A., Paruchuri, P. Persistent Safety Set Guided Offline Safe Reinforcement Learning. The Thirty-fifth International Joint Conference on Artificial Intelligence (IJCAI-2026-Main Track). (*Equal-Contribution) [**Accepted**] [[pdf](#)] [[project page](#)]

2026, **Brahmanage, J. C.**, Kumar, A. Beyond Hard Constraints: Budget-Conditioned Reachability For Safe Offline Reinforcement Learning. The Thirty-sixth International Conference on Automated Planning and Scheduling (ICAPS-2026-Planning and Learning Track). [**Accepted**] [[pdf](#)] [[project page](#)]

2025, Pham, Q. A., **Brahmanage, J. C.**, Mai, T. A., & Kumar, A. (2025, October 29). IOSTOM: Offline Imitation Learning from Observations via State Transition Occupancy Matching. The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS-2025-Main Track). [[pdf](#)])

2025, Pham, A.Q., **Brahmanage, J. C.**, Kumar, A. ShipNaviSim: Data-Driven Simulation for Real-World Maritime Navigation. The Twenty-Fourth International Conference on Autonomous Agents and Multiagent Systems (AAMAS-2025 - Main Track). [[pdf](#)]

2025, **Brahmanage, J. C.**, Ling, J., & Kumar, A. Leveraging Constraint Violation Signals For Action Constrained Reinforcement Learning. The Thirty-ninth Annual AAAI Conference on Artificial Intelligence(AAAI-2025-Main Track). (**Oral**) [[link](#)]

2023, **Brahmanage, J. C.**, Ling, J., & Kumar, A. FlowPG: Action-constrained Policy Gradient with Normalizing Flows. Thirty-seventh Conference on Neural Information Processing Systems(NeurIPS-2023-Main Track). [[pdf](#)]

2022, **Brahmanage, J.**, Kandappu, T., & Zheng, B. (2022). A data-driven approach for scheduling bus services subject to demand constraints. IEEE Transactions on Knowledge and Data Engineering (TKDE) [[pdf](#)]

2018, **Brahmanage, J.**, Ediriweera, S., Hasantha, R., Munasinghe, P., & Ranathunga, S. (2018). Annotating opinions and opinion targets in student course feedback. Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC-2018). [[pdf](#)]

2017, **Brahmanage, J.**, Ediriweera, S., Munasinghe, P., Hasantha, R., & Ranathunga, S. (2017). Opinion target extraction for student course feedback. Proceedings of the 29th Conference on Computational Linguistics and Speech Processing (ROCLING-2017), 295–307. [[pdf](#)]

Demo Papers

2023, **Brahmanage, J.**, Kandappu, T., & Zheng, B. (2023). MetroWatch: A predictive system to estimate travel attributes using smart card data. 2023 IEEE 39th International Conference on Data Engineering (ICDE-2023, Demo Paper), 3607–3610. [[link](#)]

PROJECTS

Budget Conditioned Action Pruning (*Research Project*)

2023/24

Singapore Management University

Proposed pruning based on the remaining budget during Q-value learning for safe offline RL tasks.

GFlow-RL (*Research Project*)

2023/24

Singapore Management University

Proposed the use of Generative Flow Networks to approximate Q-value distributions in high-dimensional action spaces for reinforcement learning.

Action Constrained RL with Normalizing Flows (*Research Project*)

2020/21

Singapore Management University

Proposed an algorithm that can utilize generative models to safely explore state space in Reinforcement Learning.

Travel time inference of MRT network (*Research Project*)

2020/21

Singapore Management University

Developed a solution to infer the travel time statistically within the MRT network using Automatic Fare Collection(AFC) data.

Automatic Summarization of Student Feedback (*Final Year Thesis Project*)

2017/18

University of Moratuwa.

Developed a methodology to summarize general text feedback. We developed a simple annotation scheme, a corpus, and a methodology to perform general feedback summarization.

MongoDB Support for Apache Tajo

2016

Google Summer of Code 2016 Contributor The project is to add a MongoDB storage plugin to tajo-storage.

OPEN SOURCE
CONTRIBUTIONS

React Native Floating Bubble (218+ stars)

[\[link\]](#)

Sinhala Unicode Converter (19+ stars)

[\[link\]](#)

Maritime Simulator: Data-Driven Maritime Navigation Simulation

[\[link\]](#)

ACADEMIC SERVICES **Reviewer:** AAI, NeurIPS, ICAPS, JAIR

Volunteer: GYSS 2026 (Liaison officer), SRSC 2025 (Moderator), IEEE CAI 2024 (Volunteer)

HONORS AND
AWARDS

- Dean's List -AY2026/27
- School of Computing and Information Systems, Singapore Management University.
- Awarded in recognition of significant research achievements in AY2025/26.
- Presidential Doctoral Fellowship -AY2026/27, AY2025/26, AY2024/25,
- School of Computing and Information Systems, Singapore Management University.
- NASA International Space Apps 2018
- World Finalist (best five teams in the world) representing Sri Lanka.
- Out of 25000+ participants
- Mora Ventures 2.0 (2017)
- Winner.
- Our product 'Smart Bat' became 1st out of 50 startups from University of Moratuwa
- Hack The North 2017, Canada
- World Finalist - Out of 72 countries
- IEEEExtreme Programming Competition 9.0 - Oct 2015
- National Rank 4 · Global Rank 93
- Sri Lankan Astronomy and Astrophysics Olympiad Competition
- Silver Medal
- National School Software Competition
- Merit Award
- Robotics Competition Organized by Open University Sri Lanka 2015
- **2nd** Place

FAMILIAR
PROGRAMMING
LANGUAGES AND
FRAMEWORKS

Python, C++, TypeScript, JavaScript, Haskell, Java, Pandas, NumPy, Scikit-learn, JAX, PyTorch,
TensorFlow